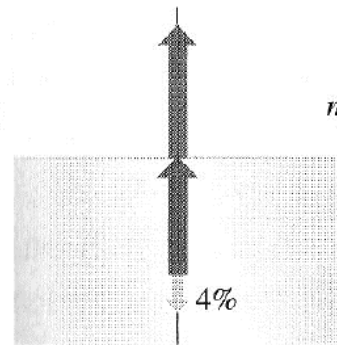


3.7 Total internal reflection

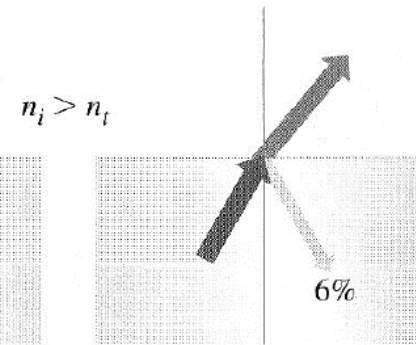
Critical angle: the special value of θ_i for which $\theta_t = 90^\circ$

$$\sin \theta_i = \frac{n_t}{n_i} \sin \theta_t$$

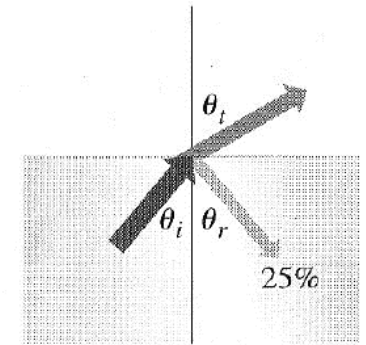
$$\sin \theta_c = \frac{n_t}{n_i} = n_{ti}$$



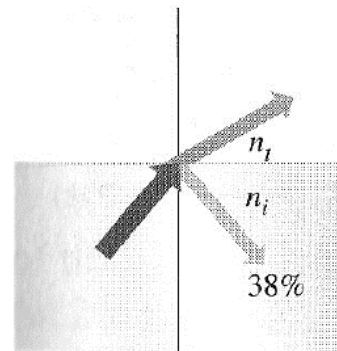
(a)



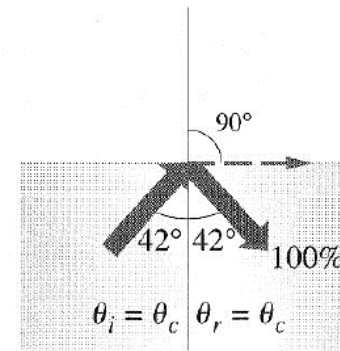
(b)



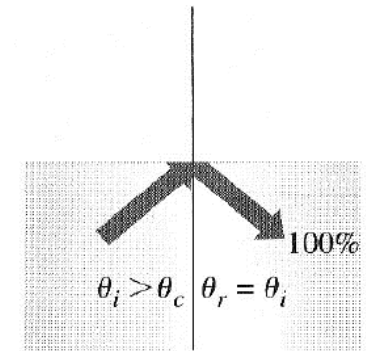
(c)



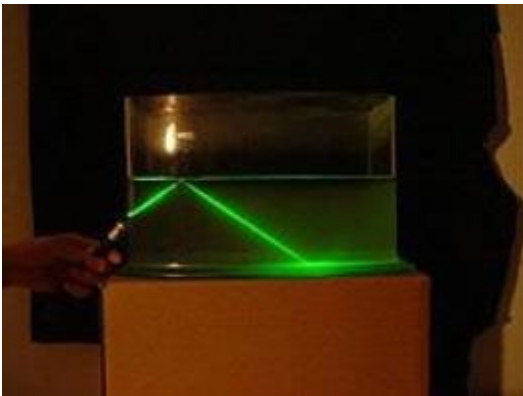
(d)



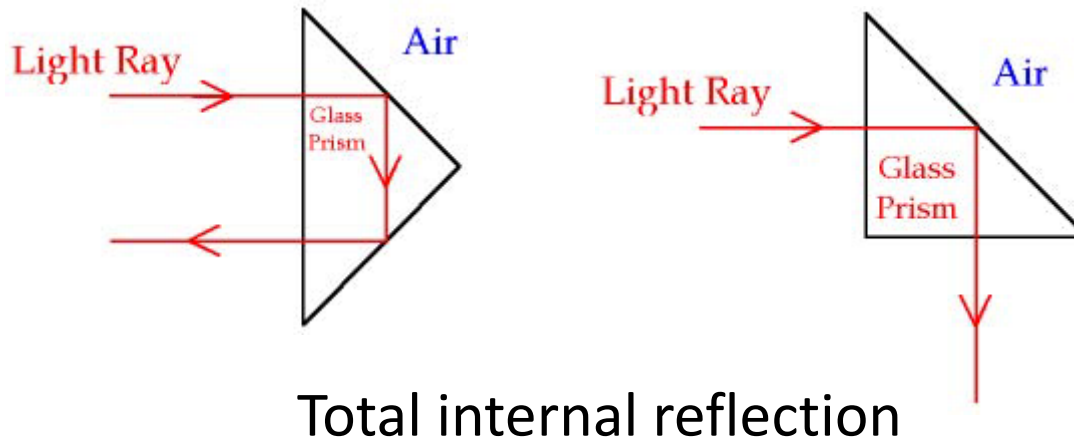
(e)



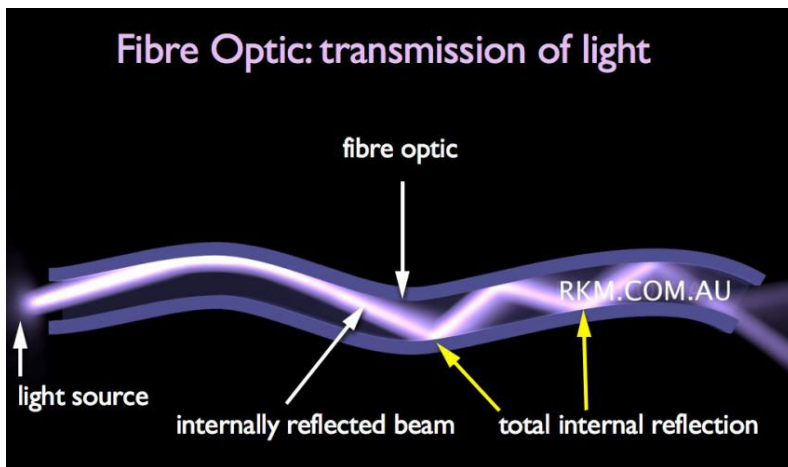
(f)



The critical angle for our air-glass interface is roughly 42° .

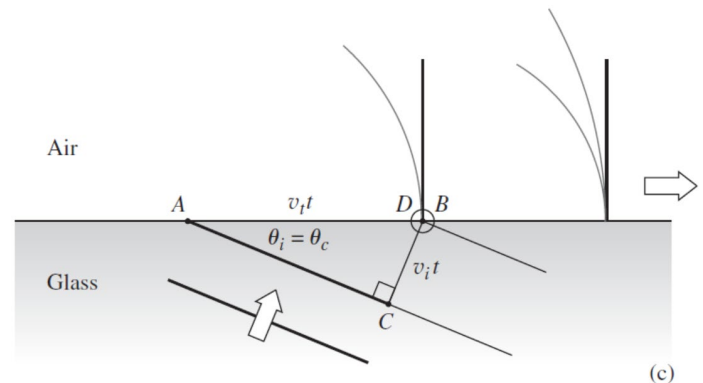
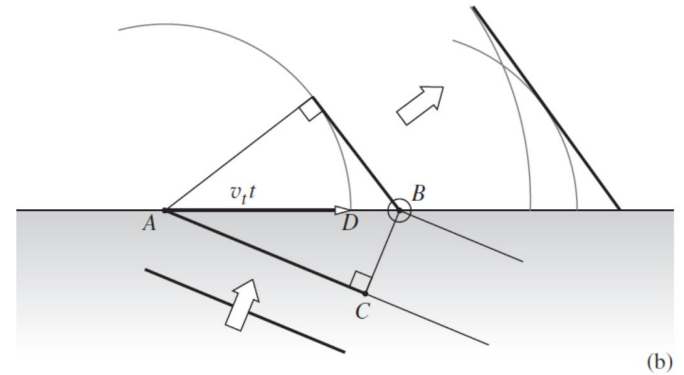
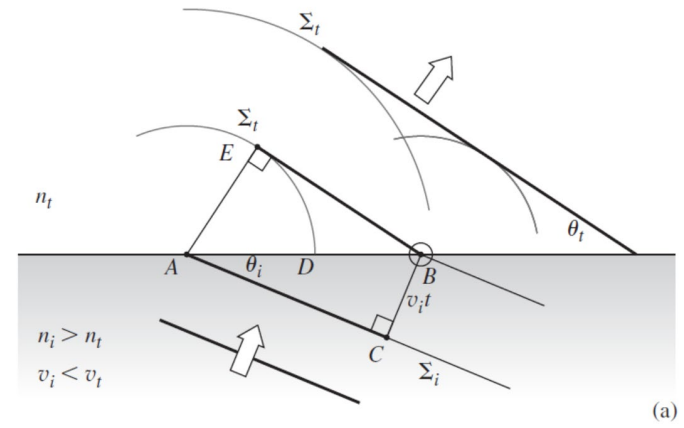


Application



Scattering perspective:

Keep θ_i and n_i constant and in successive parts of the diagram decrease n_t , thereby increasing v_t



Surface wave: for the two given media, the transmitting medium is along the interface. The resulting disturbance is known as a surface wave.

3.7.1 The evanescent wave (倏逝波)

In the case of total internal reflection: no transmitted wave.

But it becomes impossible to satisfy the boundary conditions using only the incident and reflected wave.

$$r_{\parallel} \equiv \left(\frac{E_{0r}}{E_{0i}} \right)_{\parallel} = \frac{n_t \cos \theta_i - n_i \cos \theta_t}{n_i \cos \theta_t + n_t \cos \theta_i} = \frac{n_{ti}^2 \cos \theta_i - (n_{ti}^2 - \sin^2 \theta_i)^{1/2}}{n_{ti}^2 \cos \theta_i + (n_{ti}^2 - \sin^2 \theta_i)^{1/2}}$$

$$r_{\perp} \equiv \left(\frac{E_{0r}}{E_{0i}} \right)_{\perp} = \frac{n_i \cos \theta_i - n_t \cos \theta_t}{n_i \cos \theta_i + n_t \cos \theta_t} = \frac{\cos \theta_i - (n_{ti}^2 - \sin^2 \theta_i)^{1/2}}{\cos \theta_i + (n_{ti}^2 - \sin^2 \theta_i)^{1/2}}$$

Since $\sin \theta_c = n_{ti}$, when $\theta_i > \theta_c$, $\sin \theta_i > n_{ti}$, and $r_{\parallel}, r_{\perp}$: complex quantities.

$$r_{\parallel} r_{\parallel}^* = r_{\perp} r_{\perp}^* = 1 \quad R = 1 \quad I_r = I_i \quad \text{and} \quad I_t = 0$$

Although there must be a transmitted wave, it cannot, on average, carry energy across the boundary.

The wavefunction for the transmitted electric field:

$$\vec{E}_t = \vec{E}_{0t} \exp i (\vec{k} \cdot \vec{r} - \omega t)$$

$$\vec{k} \cdot \vec{r} = k_{tx}x + k_{ty}y$$

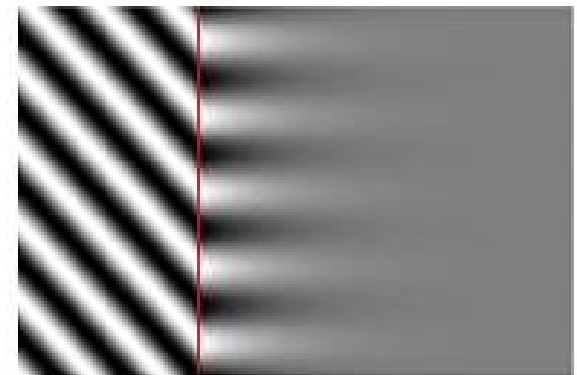
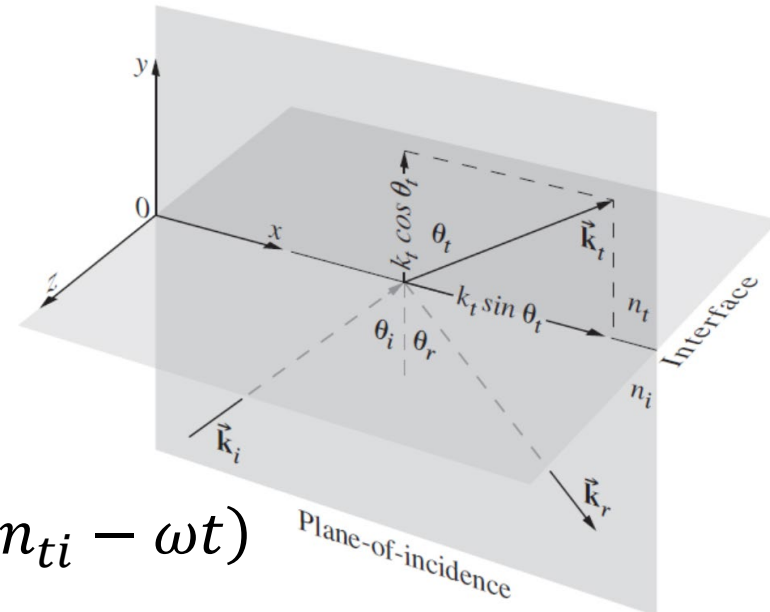
Solution:

$$\vec{E}_t = \vec{E}_{0t} \exp(-\beta y) \exp i (k_t x \sin \theta_i / n_{ti} - \omega t)$$

$$\beta = k_t \left(\frac{\sin^2 \theta_i}{n_{ti}^2} - 1 \right)^{\frac{1}{2}} \quad \delta = \frac{1}{\beta} \quad \text{penetration depth}$$

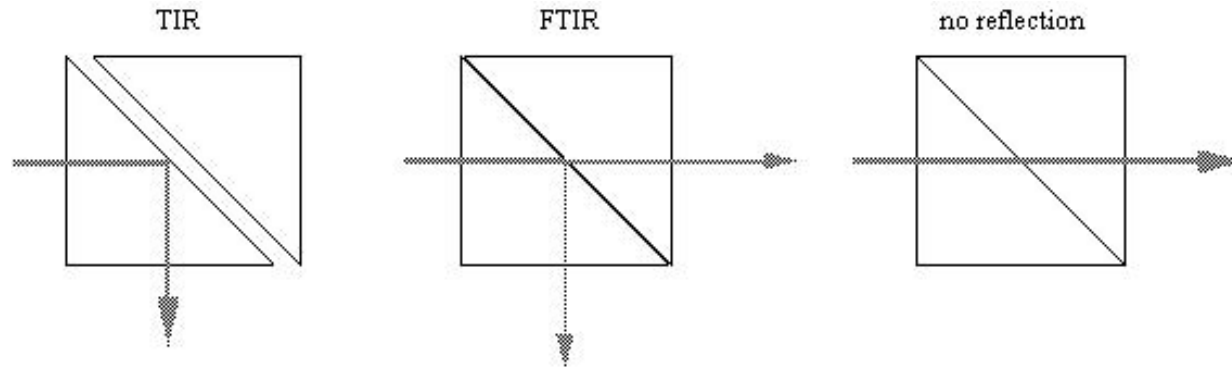
We have a wave whose amplitude drops off exponentially as it penetrates the less dense medium. The disturbance advances in the x-direction as a surface or **evanescent** wave .

Its amplitude decays rapidly in the y-direction, becoming negligible at a distance into the second medium of only a few wavelengths.

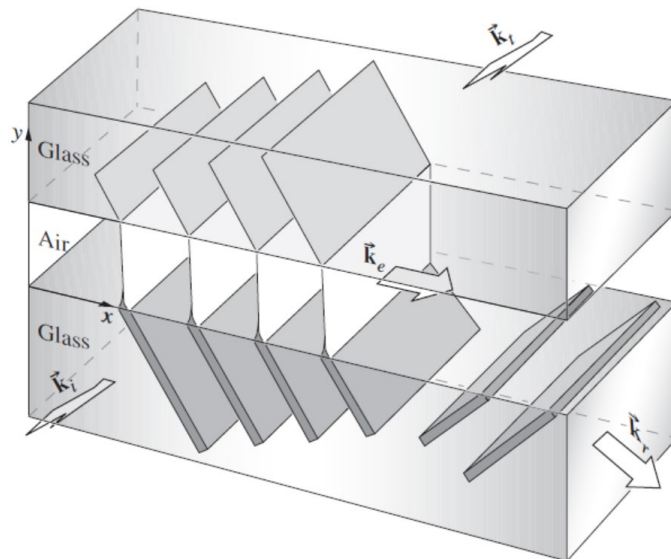


evanescent wave

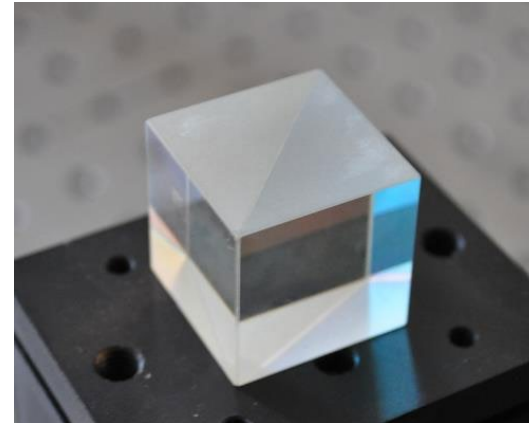
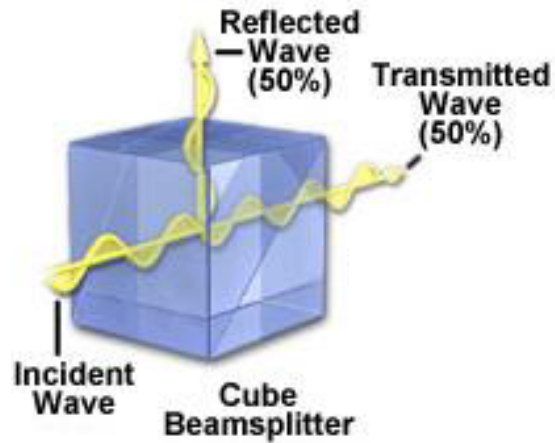
Frustrated total internal reflection (FTIR, 受抑全内反射)



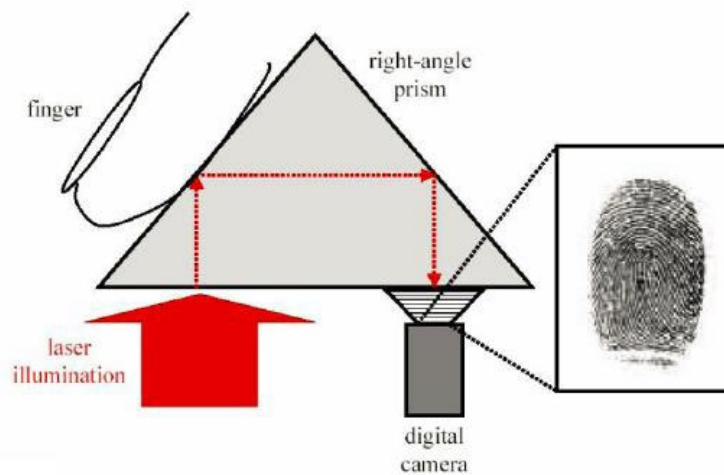
In more general terms, when the evanescent wave extends with appreciable amplitude across the rare medium into a nearby region occupied by a higher index material, energy may flow through the gap in what is known as frustrated total internal reflection (FTIR)



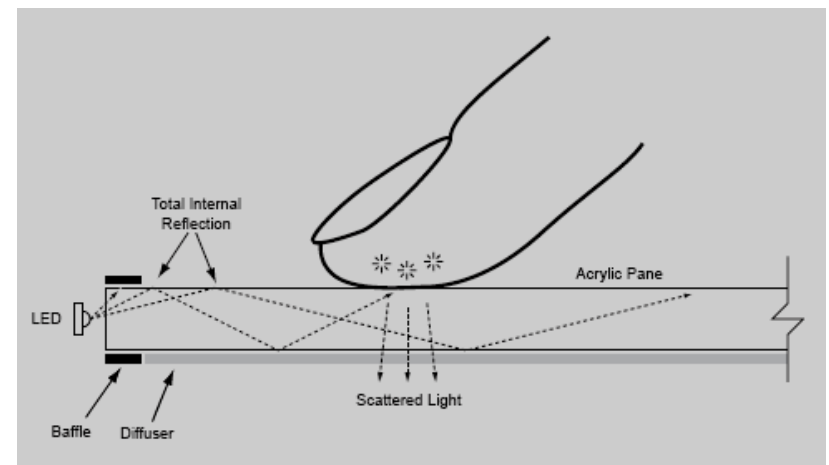
Application: Beamsplitter



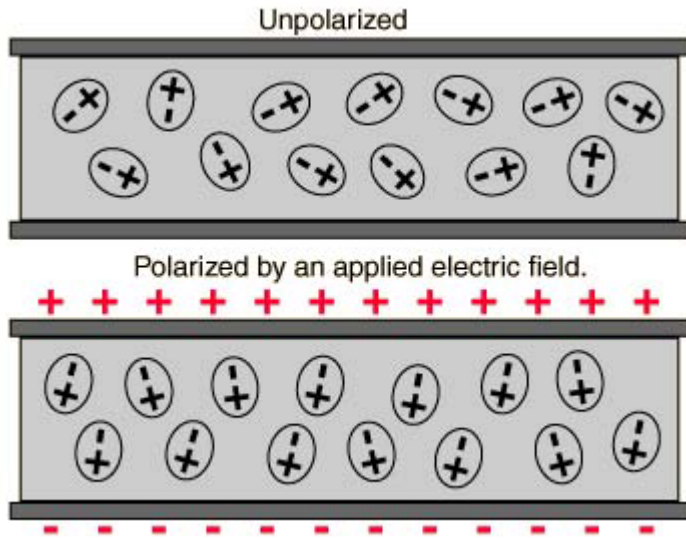
Fingerprint sensors



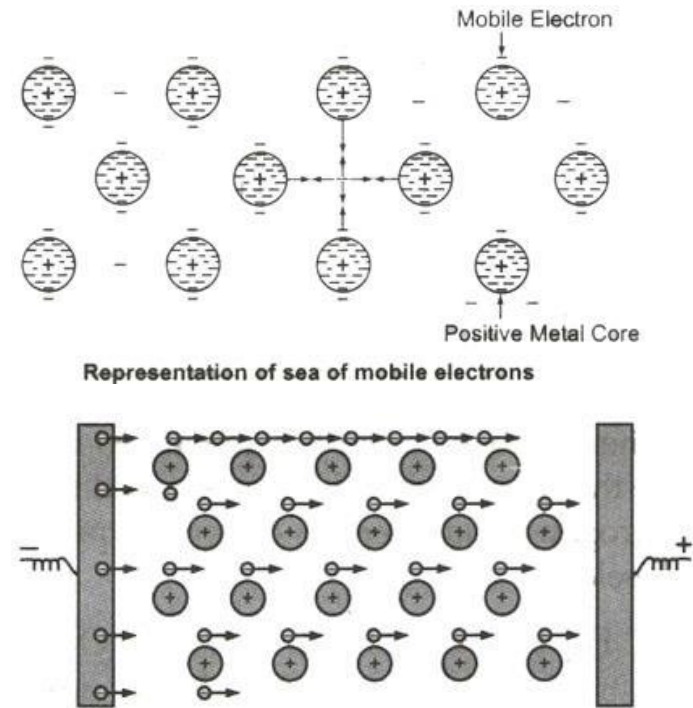
Multi-Touch Sensing through Frustrated Total Internal Reflection



3.8 Optical Properties of Metals



No free or conduction electrons
Conductivity $\sigma = 0$



Electrical conductivity in metal crystals

$$\text{Current density } \vec{j} = \sigma \vec{E}$$

The electrons, driven into oscillation by a harmonic wave, would simply follow the field's alternations. There would be no restoring force, no natural frequencies, and no absorption, only re-emission.

In real metals the conduction electrons undergo collisions with the thermally agitated lattice or with imperfections and in so doing irreversibly convert electromagnetic energy onto joule heat.

Maxwell's equations in conductors

□ Faraday's Induction Law

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d}{dt} \iint_A \vec{B} \cdot d\vec{S}$$

□ Gauss's Law- Electric

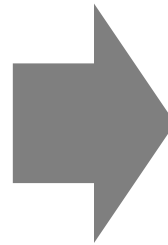
$$\oiint_A \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \iiint_V \rho dV$$

□ Gauss's Law- Magnetic

$$\oiint_A \vec{B} \cdot d\vec{S} = 0$$

□ Ampere-Maxwell's Law

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \iint_A \left(\vec{J} + \epsilon \frac{\partial \vec{E}}{\partial t} \right) \cdot d\vec{S}$$



$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\boxed{\nabla \cdot \vec{E} = \frac{\rho}{\epsilon}} = 0$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} = \mu \sigma \vec{E} + \mu \epsilon \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \cdot \vec{E} = 0$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} = \mu\sigma\vec{E} + \mu\epsilon\frac{\partial \vec{E}}{\partial t}$$

$$\left\{ \begin{array}{l} \nabla \times (\nabla \times \vec{E}) = \nabla \times \left(-\frac{\partial \vec{B}}{\partial t} \right) = -\frac{\partial}{\partial t} (\nabla \times \vec{B}) = -\mu\sigma\frac{\partial \vec{E}}{\partial t} - \mu\epsilon\frac{\partial^2 \vec{E}}{\partial t^2} \\ \nabla \times (\nabla \times \vec{E}) = \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\nabla^2 \vec{E} \end{array} \right.$$

$$\nabla^2 \vec{E} = \mu\sigma\frac{\partial \vec{E}}{\partial t} + \mu\epsilon\frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\frac{\partial^2 \vec{E}}{\partial x^2} + \frac{\partial^2 \vec{E}}{\partial y^2} + \frac{\partial^2 \vec{E}}{\partial z^2} = \mu\sigma\frac{\partial \vec{E}}{\partial t} + \mu\epsilon\frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\frac{\partial^2 \vec{E}}{\partial x^2} + \frac{\partial^2 \vec{E}}{\partial y^2} + \frac{\partial^2 \vec{E}}{\partial z^2} = \boxed{\mu\sigma \frac{\partial \vec{E}}{\partial t}} + \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

It likes the damping force in the oscillator model

The time rate-of-change of E generates a voltage, current circulate, and since the material is resistive, light is converted to thermal energy-ergo absorption.

The solution for a wave propagating in the y-direction:

$$\vec{E} = \vec{E}_0 \cos \omega(t - \tilde{n}y/c) \quad \tilde{n} = n_R - i n_I$$

$$\vec{E} = \vec{E}_0 e^{-\omega n_I y/c} \cos \omega(t - n_R y/c)$$

The disturbance advances in the y-direction with a speed c/n_R , its amplitude is exponentially attenuated.

Irradiance

Absorption coefficient: $\alpha \equiv 2\omega n_I/c$

$$I(y) = I_0 e^{-\alpha y}$$

Skin depth: $y \equiv 1/\alpha$

Example: copper

For ultraviolet wavelength ($\lambda_0 = 100 \text{ nm}$): Skin depth $y = 0.6 \text{ nm}$

For infrared wavelength ($\lambda_0 = 10000 \text{ nm}$): Skin depth $y = 6 \text{ nm}$

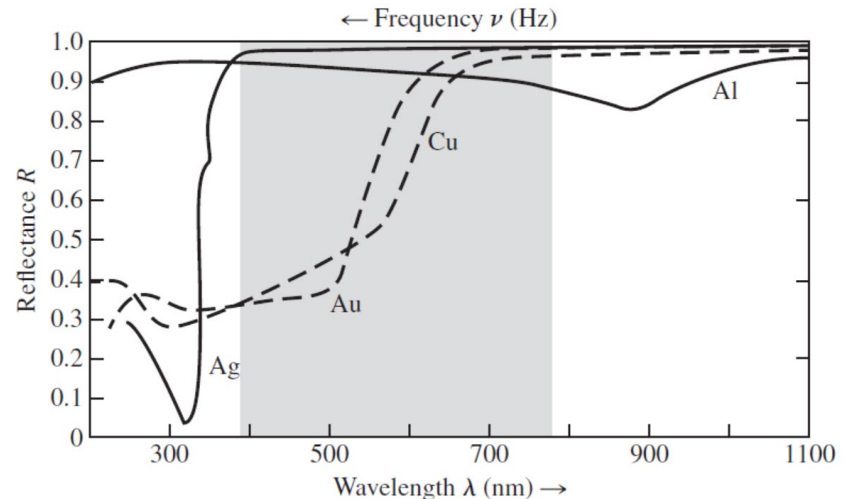
The familiar metallic sheen of conductors corresponds to a high reflectance, which exists because the incident wave cannot effectively penetrate the material.



Relatively few electrons in the metal “see” the transmitted wave, and therefore, although each absorbs strongly, little total energy is dissipated by them. Instead, most of the incoming energy reappears as the reflected wave.

For the simplest case of normal incidence on a metal

$$R = \frac{I_r}{I_i} = \left(\frac{\tilde{n} - 1}{\tilde{n} + 1} \right) \left(\frac{\tilde{n} - 1}{\tilde{n} + 1} \right)^* \\ = \frac{(n_R - 1)^2 + n_I^2}{(n_R + 1)^2 + n_I^2}$$



Homework

□ 4.14, 4.39, 4.50, 4.66, 4.70, 4.72, 4.76